

ScanTailor Spectre



Version 2.0a34 | macOS (Apple Silicon) | Requires macOS 15 or later

ScanTailor Spectre transforms raw scans into clean, publication-ready pages. Import a PDF or folder of images, process through a 10-stage workflow, and export a polished, searchable PDF.

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Version 2.0a34 — The Second Speed Build

Released July 24, 2026. Version 2.0a34 is the second major performance release for modern Apple Silicon Macs. On the same 392-page scanned-book workflow, observed end-to-end times progressed from **6 m 38 s in v2.0a19** to **3 m 28 s in v2.0a33**, then to the following choices in v2.0a34:

OCR setting	392-page time	Intended use
Accurate OCR + language correction	2 m 48 s	Highest measured text quality
Accurate OCR + skip language correction	1 m 30 s	Best speed/quality balance
Fast OCR + skip language correction	29.5 s	Maximum throughput; largest OCR accuracy tradeoff

These are measured runs on the reference book, not a promise that every book or Mac will finish in exactly the same time. Fast OCR is substantially less accurate than Accurate OCR; use the 1 m 30 s setting when searchable-text quality still matters.

- Auto Process now keeps decoded PDF pages in a RAM-aware cache and coalesces concurrent requests for the same page, avoiding repeated rasterization. It also combines compatible pipeline stages and carries Output's newly processed in-memory image directly into OCR instead of saving and reloading it or running Output twice.

- Mixed mode is now conservatively autodetected from the existing Finalize color analysis. It finds localized continuous-tone pictures on otherwise document-like pages without adding another full-image detection pass, and the choice is preserved correctly through Finalize and Output.
- The OCR stage and Auto Process dialog now expose the same three remembered controls: **Fast OCR**, **Multilingual detection**, and **Skip language correction**.
- The language choices are now honest about their cost. **System language** is the fast default; **Multilingual detection** enables true automatic language detection and is deliberately labeled slower. A selected primary language is used only when multilingual detection is off.
- OCR result caching now includes the recognition mode, language-correction choice, engine schema, and output-file identity, so changing OCR settings or regenerating a page cannot silently reuse stale text.
- Fixed unbounded Apple Vision object lifetime growth during long OCR runs and retained stable page-parallel Vision OCR as the default.
- macOS 26's new document-recognition engine was investigated using only Weasel's newly processed in-memory page images. It can be faster on short runs, but concurrent requests crashed inside Apple's framework during full-book stress tests, while the stable serialized form was slower than the normal engine. It therefore remains research-only and opt-in through `SCANTAILOR_DOCUMENT_VISION_OCR=1`; it never reads or reuses text from the original PDF.
- Added performance instrumentation and regression coverage for image-cache behavior, OCR cache validity, color/Mixed classification, PDF loading, and long-run stability. The release keeps the crash-prone document-OCR concurrency path disabled.

Version 2.0a33 — The First Speed Build

Measured speedup: a 392-page scanned book (*The Best Short Stories of 1924*, [Internet Archive](#)) auto-processes end-to-end — including accurate-mode OCR — in **3 m 28 s (0.53 s/page)**, versus 6 m 38 s on v2.0a19: about **1.9× faster overall, and 5–6× faster in the processing pipeline itself** (OCR time is Apple Vision's and unchanged). Color classification on the same book improved from 340 pages wrongly detected as Color to 388 correct B&W pages, with only the covers (grayscale) and the bookplate/library card (color) treated specially.

- Major performance overhaul: parallel PDF rasterization, decoded-image cache, coalesced thumbnail updates, batch queue refill before UI work, gated per-page parallelism.
- OCR receives the rendered output image in memory; PDF export moved off the GUI thread with cancellation and atomic file replacement.
- New unified “Auto Process...” dialog: color handling preset (Force B&W / B&W + grayscale / Best guess), OCR on/off, remembered settings.
- Auto-process runs report elapsed time and per-stage breakdown.
- Fixed three long-standing silent image-pipeline bugs: gray morphology min/max was inverted in the accelerated (vImage/Metal) paths since December, distorting Mixed-mode picture detection and page-split line finding; grayscale downscaling used a resampler that strayed up to 66 gray levels from the intended area average; and morphological reconstruction (seed fill) could stop propagating early, subtly affecting picture detection and background estimation.
- Added acceleration parity tests so GPU/SIMD paths can no longer silently diverge from the reference implementation; test suites are now fully green.

- Finalize no longer silently misclassifies aged/toned paper as Color when margin sampling fails.

Version 2.0a32

- PDF export now opens in the source document's folder with a safe `<source>_processed.pdf` fallback name.
- Saving a new project now carries already-generated pages and thumbnails from temporary storage into the project's output folder.

Version 2.0a31

- Export now includes book metadata fields, recommended PDF filenames, ISBN lookup, and optional send-to-Zotero support for the locally running Zotero app.
- After exporting a PDF, the success dialog now has a **Reveal in Finder** button so you can jump straight to the file you just saved.
- Margins now supports full-bleed pages with the F shortcut; full-bleed pages ignore content boxes and match-size aggregation.
- The Margins stage now has one **Apply To** flow for page layout instead of separate competing margin/alignment buttons.
- The macOS startup view no longer falls back to the oversized legacy gray project panel.
- Output uses the native controls again to avoid QtWebEngine/macOS accessibility crashes.

Version 2.0a30

- Output photo adjustments now show a single, unified panel for Color and Grayscale pages. No more duplicated panels, and no empty gap between the sliders and Dewarping.
- Sliders feel smoother during drag: the app no longer stutters while you fine-tune adjustments.
- Exposure, Contrast, Highlights/Shadows, and Whites/Blacks now apply in the same order Lightroom uses, so results match what you'd expect from other photo tools.
- Stage 2 (Split Pages): opening older projects no longer re-runs detection on every page.
- Stage 2 (Split Pages): art books with broad, dark binding gutters are detected more reliably.
- Stage 4 (Page Box): applying a page box across pages of different sizes or rotations now stays centered instead of drifting off the page.
- Dark-theme slider handles now render cleanly (previously slightly clipped).
- Stability and packaging improvements: the app bundle is more self-contained and safer to close during the Output stage.

Version 2.0a29

- Output photo-adjustment sliders are now responsive: thumbnails redraw once after a 300 ms pause instead of on every drag tick.

- Clicking anywhere on a slider track now jumps the handle to that position (Lightroom-style), in addition to the existing drag behaviour.
- Numeric value boxes next to each slider are now compact and no longer expand to fill the column.
- Removes a stale snap-to-zero behaviour that interrupted slider drags mid-stroke.

Version 2.0a28

- Adds photo adjustment controls in Output, including exposure, contrast, highlights, shadows, whites, blacks, temperature, and tint.
- Fixes photo adjustments so they apply properly to multiple selected pages and refresh the thumbnails afterward.
- Fixes Output so B&W and grayscale pages show the right controls instead of the full color adjustment panel.
- Sets the minimum supported macOS version to 15.

Features

- **Auto Mode** - One-button processing through all stages with smart defaults: majority-vote page splitting, zero deskew, content outlier detection, and size outlier handling
 - **Apple Silicon Native** - Gatekeeper-friendly application
 - **PDF Import** - Open PDFs directly, no need to extract pages first
 - **Batch Processing Summaries** - Dialogs after stages 2, 4, 5 to catch problems and jump to pages needing attention
 - **Detection Settings** - Adjustable Fill Factor and Border Tolerance for art books and photo-heavy content
 - **Finalize Stage** - New stage for color mode selection and output format
 - **Intelligent Color Detection** - Auto-detects B&W vs grayscale vs color pages, including embedded photographs
 - **Photo Adjustments** - Temp, Tint, Exposure, Contrast, Highlights, Shadows, Whites, Blacks sliders with Auto and Reset
 - **Packaged Web Output Panel** - Output stage web controls now ship with the required Qt WebEngine helper dependencies in the app bundle
 - **OCR Stage** - Automatic text recognition for searchable PDFs
 - **Export Stage** - New dedicated PDF export with quality presets
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Quick Start

1. Import Your Scans

From the startup screen or from the file menu: - **Import PDF** - Extract pages from an existing PDF - **Import Folder** - Load a folder of scanned images - **Import Project** - Load a project you have previously saved.

Supported formats: PDF, TIFF, PNG, JPEG, BMP

Recently saved projects will be shown here.

2. Work Through the Filters

Each filter stage refines your pages. Work top to bottom:

#	Stage	Purpose
1	Fix Orientation	Rotate pages right-side up
2	Split Pages	Separate two-page spreads
3	Deskew	Straighten tilted scans
4	Page Box	Define page boundaries
5	Select Content	Crop to content area
6	Margins	Set page margins
7	Finalize	Choose B&W, Grayscale, or Color
8	Output	Apply image processing
9	OCR	Text recognition for searchable PDFs
10	Export	Create final PDF

To run a stage, click on the little triangular play button next to the stage you are at. Stage 10 is special. You don't need to run it, you simply click on Export to PDF.

3. Auto Mode or Batch Process

Auto Mode (Cmd+Shift+A) processes all pages through stages 2-8 with smart defaults. It automatically detects majority page layout, sets deskew to zero, preserves layout for low-coverage pages, and detaches size outliers from aggregate sizing. For OCR as well, use Cmd+Shift+O. You can stop auto mode at any point with the stop button.

For manual control, batch process individual stages using the play button. If you need to split spreads to pages, run stage 2. If not, or if you have run stage 2 already go to stage 6, Finalize, and run it and each of the next stages.

After batch processing certain stages, a summary dialog appears to help catch common problems:

- **Stage 2 (Split Pages)** - Shows pages that were/weren't split, lets you force changes

- **Stage 4 (Select Content)** - Shows pages with low content coverage that might need layout preserved
- **Stage 5 (Margins)** - Warns about unsplit spreads or outlier pages that cause margin issues. If you have unspilt spreads, you will have to go back to Stage 2.

The dialogs in stage 2 and 5 let you jump directly to problem pages and take corrective action without hunting through hundreds of thumbnails.

The Workflow Explained

Each stage has automatic operations and potentially requires minimal input, but some scans can be more difficult to process than others. For example, a decent scan with only text pages might require virtually no input on your part. A 500 page book with color pages, some of which are skewed, many of which have color casts and require individual adjustments may require substantial intervention.

Although ScanTailor Spectre is designed to be as automated as possible, it can make mistakes. It's always best to check your output.

After you run a stage, the parameters that stage set are stored in your project on a per-page basis. Let's say you set a page to color once. If you rerun the automatic color detection, it will not alter that setting even if it thinks it is grayscale.

This is pre-beta software. Save your project frequently.

Before a new project is saved, generated page images and thumbnails are kept in macOS temporary storage. **Save Project** creates a self-contained project folder and moves a permanent copy into its output subfolder; closing an unsaved project may remove the temporary copy. The final PDF is created only when you use **Export to PDF**.

Stage 1: Fix Orientation

Corrects pages that are upside-down or rotated. Auto-detection handles most cases; use the rotate buttons for manual override. It is rare that you will have to even run this separately unless your scans are awful. This is not the same as Stage 3, Deskew.

Stage 2: Split Pages

Separates book spreads (two pages per scan) into individual pages. The split line is auto-detected but can be adjusted manually.

The current detector is substantially different from Version 2.0a24. In 2.0a24, page splitting mostly relied on the older geometry/Vision decision path and then trusted the center-ish split it found. That worked for clean text spreads, but it was weak on art books and photo-heavy scans where the visible gutter is a broad shadow rather than a thin white gap.

In the current 2.0a25 line, Split Pages combines Apple Vision text-region evidence with a dedicated spine-darkness search. Vision provides a coarse anchor and page-number evidence; SpineDarknessFinder then tests candidate gutter columns using vertical persistence, local darkness, neighbor contrast, and several narrow rescue paths for real

binding shadows next to photos or text. The goal is not to make every spread automatic, but to reduce the number of obvious manual fixes in books with photographs, illustrations, and asymmetric page layouts.

Recent detector work added: - page-number-aware refinement, so two detected page numbers count as strong evidence for a two-page spread - conservative handling when only one page number is found - broad dark-gutter handling, including protection against snapping to the wrong edge of a gutter band - a photo-left/text-right rescue for spreads where the true gutter is dark and both neighboring bands are also dark, so the old “one neighbor must be blank paper” rule would reject it - no-gutter regression guards, so light text-only spreads are not forced into the photo/book-gutter rescue paths

The detector can still be wrong. For difficult art books, inspect Stage 2 before continuing, especially spreads with full-page photographs, chapter openers, or images that run into the gutter.

After batch processing, a summary dialog appears with two toggle buttons: - **Not Split (X)** - Pages kept as single pages (may need to be split) - **Split (Y)** - Pages that were split (may have been split incorrectly)

Double-click any page to jump to it for inspection. Use the action buttons to force changes: - In “Not Split” view: **Force Two-Page** makes pages split - In “Split” view: **Force Single Page** makes pages unsplit

Visual indicators in the list: - Gray strikethrough = page was visited/jumped to - Dark green strikethrough = action was taken (force split or unsplit)

Stage 3: Deskew

Straightens pages that were slightly tilted during scanning. Even 1-2° of tilt is noticeable in output.

Tip: Sort by “decreasing deviation” to review the most-skewed pages first. The sort order is located at the bottom of the thumbnail panel on the right.

Thumbnail Filters: The B, G, C buttons at the bottom of the thumbnail panel filter pages by color mode (B&W, Grayscale, Color). Click to show only pages of that type—useful for reviewing all color pages at once or focusing on B&W pages that need adjustment.

Stage 4: Page Box

Defines the physical page boundaries within the scan. This is useful when the scanner captured more than just the page (scanner bed edges, background), or when you want to define a consistent page size across all scans.

Page Box options: - **Disable** - Don’t detect page boundaries; use the entire image - **Auto** - Automatically detect the page edges (looks for the transition from scanner background to paper) - **Manual** - Draw the page boundary yourself by dragging corners and edges

When Auto mode is selected, **Fine Tune Page Corners** runs a second pass after initial edge detection. It steps each corner diagonally inward, skipping over dark pixels (shadows, bent page edges) until it finds light pixels (actual paper). This is useful for thick books where page corners cast shadows or curl away from the scanner bed.

In Manual mode, you can enter exact **Width** and **Height** values to set a specific page size.

Sorting: Use “Order by increasing width” or “Order by increasing height” to quickly find pages with incorrect page boundaries. This is the key advantage of having Page Box as its own stage — you can review and fix page boundaries before running content detection.

Stage 5: Select Content

Defines the **Content Box** — the area containing actual content like text and images. Everything outside this box becomes white margin in the final output. The content box is constrained within the page box set in the previous stage.

When do you need this? The Content Box identifies where text or images are so margins can be standardized. For most workflows, Auto mode handles this well, but you may need Manual adjustments for pages with unusual layouts (title pages, illustrations, fold-outs) or when auto-detection picks up shadows or artifacts as content.

Content Box options: - **Disable** - Don’t detect content; use the entire page - **Auto** - Automatically find text and images on the page - **Manual** - Draw the content area yourself

Detection Settings

When using Auto mode for Content Box, these settings control how content is detected:

- **Fill Factor** (0.50-1.00, default 0.65) - Controls what density of content is accepted. The default is optimized for text documents. For art books, photo-heavy pages, or pages with large dark areas, increase this value:
 - **0.65** - Default, works well for text documents
 - **0.85** - Better for mixed text/image content
 - **0.95+** - Use the entire page as content (best for art books, covers, full-page images)
- **Border Tolerance** (0-20px, default 2) - Controls how much content touching page edges is preserved:
 - **Low (0-5)** - Aggressively trim edge content (removes shadows, scanner artifacts)
 - **High (15-20)** - Preserve content that extends to page edges (for images that bleed to margins)

Tip: For art books or pages with large photographs, set Fill Factor to 0.95+ to capture the entire page content. The default settings are optimized for text documents and may clip images.

This excludes: - Scanner bed edges - Book margins you want to remove - Fingers or page holders

Auto detection is not always perfect, especially with images or decorative elements. Sort by “Order by completeness” to review pages that may need manual adjustment.

Content Coverage Summary

After batch processing, a summary dialog appears showing pages grouped by content coverage: - **Low Coverage** - Pages where detected content covers less than the threshold percentage of the page area. These may have intentional layout (chapter titles, short paragraphs) that shouldn't be centered. - **Normal Coverage** - Pages where content fills most of the page area.

Use the **Coverage threshold** slider (20%-80%) to adjust what counts as "low coverage."

Actions for low coverage pages: - **Jump to Selected** - Navigate to a page for manual inspection - **Preserve Layout (Selected/All)** - Disable content detection for these pages, keeping their original layout intact

Visual indicators: - Gray strikethrough = page was visited/jumped to - Dark green strikethrough = action was taken (preserve layout)

Stage 6: Margins

Sets white space around content in the final output and page size.

- **Top/Bottom/Left/Right** - Individual margin sizes
- **Alignment** - Where content sits within the page

Stage 7: Finalize

This stage determines how each page will be processed:

- **Color Mode:** Black & White, Grayscale, or Color
- **Output Format:** TIFF or PNG (you can ignore this if you don't care to save the individual images)
- **Output Location:** Where processed files are saved

The app auto-detects the appropriate color mode. Obviously, Color pages require the most storage space, Grayscale requires less, and Black and White the least of all.

The **Midtone Threshold** slider adjusts detection sensitivity, useful for smaller images (lower = more pages will be inspected and detection will be slower). You can override the determination after inspection.

White Balance (Color and Grayscale modes): - **Force white balance** - Automatically corrects color casts from aged paper or poor lighting - **Pick Paper Color** - Click this, then click on an area that should be white/neutral in the preview. The app will correct the entire page based on that sample.

Clear All Pages resets all pages to unprocessed state, useful if you want to re-run automatic detection with different settings.

Previous versions of ScanTailor exported to images, the workflow in this one is images or PDF to PDF. Images will be discarded when you close the project unless you choose to preserve them at this point by ticking the box to "Preserve Output Images."

Stage 8: Output

Applies image processing to generate the output files. Options vary by color mode.

Output Resolution: Sets the DPI of the output image. 400 DPI is the Library of Congress recommendation for most documents. Higher = sharper but larger files.

Black & White Mode Options

- **Binarization Threshold** - Controls the cutoff between black and white. Lower values = more black, higher = more white. Adjust if text appears too thin or too bold.
- **Normalize Illumination** - Evens out lighting variations across the page. Helps with scans that have shadows or uneven exposure.
- **Morphological Smoothing** - Smooths jagged edges on text and lines.
- **Morphological Opening/Closing** - Advanced cleanup options for removing small artifacts.

Grayscale & Color Mode Options

- **Normalize Illumination** - Evens out lighting variations.
- **Sharpen/Blur filters** - Enhance or soften the image.

Pass-through Mode

Enable **Pass-through** to skip heavy processing (binarization, despeckling, dewarping) while still applying adjustments. Useful for: - Pre-processed pages that only need brightness/contrast tweaks - Covers and illustrations that should remain untouched except for tonal adjustments - Pages where automatic processing produces poor results

When pass-through is enabled, the page is scaled to output DPI and adjustments (brightness, contrast, auto levels) are applied, but no other processing occurs.

Brightness & Contrast

- **Brightness** - Adjusts overall image lightness. Slide right to brighten, left to darken.
- **Contrast** - Adjusts tonal range. Slide right to increase contrast, left to decrease.
- **Auto Levels** - Stretches the histogram to use the full 0-255 range for maximum contrast.

The sliders have tick marks at -100, 0, and +100, with a center detent that snaps to 0 for easy reset. These adjustments apply even when pass-through mode is enabled.

Paper Detection (Equalize Illumination)

When “Equalize illumination (Color)” is enabled, these advanced controls appear:

- **Min Brightness** (0-255, default 120) - Pixels brighter than this may be paper
- **Max Saturation** (0-255, default 60) - Pixels less saturated than this may be paper
- **Min Coverage** (0-50%, default 1%) - Minimum paper area required for equalization
- **Adaptive (sample margins)** - Auto-detect paper color from page margins

These controls help the algorithm distinguish paper from content when correcting uneven lighting.

Common Options (All Modes)

- **Dewarping** - Flattens curved pages from book spines. Set to “Auto” for automatic detection or “Manual” to adjust control points yourself. Most useful for thick books where pages curve near the spine.
- **Despeckle** - Removes small dots and noise:
 - *Cautious* - Minimal cleanup, preserves detail
 - *Normal* - Balanced approach
 - *Aggressive* - Heavy cleanup, may remove fine details
- **Equalize Illumination** - Additional lighting correction.
- **White Margins** - Ensures margins are pure white.

Zones

- **Picture Zones** - Draw rectangles around photographs or illustrations. These areas are processed differently to preserve detail and gradients instead of being binarized.
- **Fill Zones** - Draw areas to fill with a solid color (usually white). Useful for removing stamps, stains, or unwanted marks.

To draw a zone, select the zone tool, then click and drag on the preview image.

Apply To...

Once you’ve tuned settings for one page, use **Apply To...** to copy those settings to other pages. You can apply to: - All pages - Selected pages only - Pages with the same color mode

This is essential for efficiently processing large documents.

Stage 9: OCR

Performs optical character recognition to make your PDF searchable. When enabled, text is recognized and embedded as an invisible layer in the exported PDF, allowing you to search and select text.

Options: - **Enable OCR** - Toggle text recognition on/off (enabled by default) - **Language** - Select the document language for better recognition accuracy, or leave on Auto-detect - **Accurate Recognition** - Use accurate mode for best results (slower) or fast mode for quicker processing

Status Display: - Shows whether the current page has been processed - Displays the number of text blocks found

OCR runs automatically when you batch process through the Export stage. Results are cached per-page, so re-running won’t re-process pages that already have OCR data unless you clear them.

Stage 10: Export

Creates the final PDF.

Max DPI: Limits output resolution for grayscale and color pages.

400 DPI is the Library of Congress recommendation for most documents. If your scan is lower quality or to be read on screen only, you may find that lowering the resolution as low as 72 DPI gives acceptable results and a much lower final PDF size. At 72 DPI, however, it would be very difficult to read black and white pages, which is why they are controlled by the master output resolution in Stage 7.

Keyboard Shortcuts

Key	Action
Cmd+Shift+A	Auto Process All (stages 2-7)
Cmd+Shift+0	Auto Process All + OCR (stages 2-8)
Page Up/Down	Previous/next page
Home / End	First/last page
Cmd+S	Save project

Finalize (7) & Output (8) Stage Shortcuts:

Key	Action
C	Set selected page(s) to Color mode
G	Set selected page(s) to Grayscale mode
B	Set selected page(s) to Black & White mode
Shift+C	Toggle Color page filter
Shift+G	Toggle Grayscale page filter
Shift+B	Toggle B&W page filter

Output (8) Stage Only:

Key	Action
P	Toggle Pass-through mode

Tips

Scanning

- **300-400 DPI** for text documents
- **600 DPI** for fine detail or small text
- Scan in **color** even for B&W content - better conversion results

Processing

- Work through stages **in order** - each depends on the previous
- Configure **one page well**, then batch apply to similar pages
- Use **page ordering** options to find problem pages
- **Save frequently** - all settings are preserved in the project file

Color Mode Selection

- **B&W (1-bit)**: Smallest files, pure black and white, ideal for text-only
- **Grayscale**: Better for pages with photos or illustrations
- **Color**: Only when color information matters

Typical File Sizes (100-page book)

- B&W PDF: 5-15 MB
- Grayscale PDF: 20-50 MB
- Color PDF: 50-150 MB

Project Files

Save Project creates a self-contained folder containing:

- The .ScanTailor project file
- Copies of the source images in *originals*
- Generated page images and thumbnails in *output*
- All settings and processing state

The exported PDF is separate. Create it from Stage 10 with **Export to PDF**.

Credits

ScanTailor Spectre is based on: - [ScanTailor Advanced](#) by 4lex4 - [ScanTailor](#) by Joseph Artsimovich

License

GPL-3.0 - See [LICENSE](#) for details.

ScanTailor What?

ScanTailor Spectre is a Mac-native fork of ScanTailor Advanced, developed by Claude. Several considerations informed the choice of its name.

First, the name serves to distinguish it from other community releases, including ScanTailor, ScanTailor Advanced, and ScanTailor Experimental.

Second, the title references the opening line of the Communist Manifesto: “A spectre is haunting Europe.” Jacques Derrida, in “Spectres of Marx” wrote that the dead refuse to remain absent. For Derrida, this refers to the spectre of Marxism, but it can also refer to the haunting of the Internet by the ongoing persistence of ScanTailor as well as to the scanned book existing as a ghostly trace circulating online rather than as a physical object.

Fourth, ScanTailor Spectre is developed using Claude Code. The name also draws attention to the role of unauthorized book scans in the development of large language model artificial intelligences. It is illegal to copy and distribute copyrighted books, and we do not condone using ScanTailor Spectre for such purposes. Both Anthropic and Meta trained their models on millions of books obtained from shadow libraries such as Library Genesis, which are motivated by a vision of information freedom championed by Aaron Swartz. Recently, Anthropic settled with authors for \$1.5 billion, equating to approximately \$3,000 per book, a sum that exceeds the lifetime earnings of most books. In contemporary society, intellectual property is controlled not by individuals, but by those in positions of power.

Lastly, the first application I vibe-coded was a clone of the 1990s game Spectre VR.

The “spectre in the machine” draws on all of these.